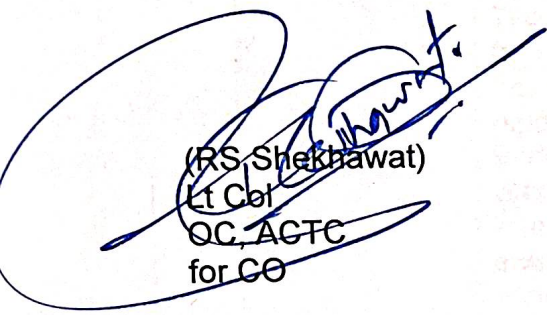


Appx A

(Refers to Para 2 of S of C : Procure & Installation of Digital Language Lab)

LIST OF ITEMS FOR PROCUREMENT AND INSTALLATION OF ONE DIGITAL LANGUAGE LAB WITH COMPLETE ACCESSORIES FOR ACTC, NO 2 TRG BN, ASC CENTRE (SOUTH) OUT OF INFORMATION TECHNOLOGY HARDWARE FOR THE FY 2026-27

S.No	Description	A/U	Qty Reqd	Remarks
1.	Audio Visual Tools for English	Nos	41	Detailed specification is att as Annexure-I
2.	High Quality and Rugged Headset with Mic	Nos	41	Detailed specification is att as Annexure-II
3.	Desktop PC All-in-One Intel Core i7 13 th Gen or higher/ 16 GB RAM/ 512 GB SSD/Win 11 Pro/ 23.8" FHD/ 3yrs onsite warranty	Nos	41	Detailed specification is att as Annexure-III
4.	Online UPS 20 KVA with 60 minutes backup	Nos	01	Detailed specification is att as Annexure-IV
5.	Network Components with installation and commissioning of Lab work	Work	01	Detailed specification is att as Annexure-V


(RS Shekhawat)
Lt Col
QC, ACTC
for CO

Station : Bangalore -07

Date : 08 May 2026

Annexure-I

MINIMUM REQUIRED TECHNICAL SPECIFICATIONS : FOR PROCUREMENT & INSTALLATION OF ONE DIGITAL LANGUAGE LAB WITH COMPLETE ACCESSORIES

TECHNICAL SPECIFICATION : AUDIO VISUAL TOOLS FOR ENGLISH			
Ser No	Specification	Description	Remarks
1.		The Digital Language Lab Control / communication Systems/ Module. The leading system includes all the features and functions that are essential for effective and enjoyable teaching and learning, such as preset language learning activities, screen transfer features, audio and text communication tools, web browsing and program launch capabilities, file functions and classroom management and control features.	
		The students can be divided into six different sessions. The sessions can be controlled separately and can simultaneously perform different activities.	
		The system should provide the teacher to control the whole class with different controlling toll like Lock students screen, lock students keyboard and mouse, take remote control of student computer, shutdown student computer	
		The teacher should be able to send the instructor's screen, audio, video and text based content to student's consoles for teacher presentations or broadcast any students screen to other students.	
		The teacher should be able to disable the student's stations from the teacher's station control panel.	
		Teacher and students should be able to interact (speaking/listening) using headset with built-in microphone. Either for a single student or a group of student or whole class.	
		The system should have a media library area which will enable a teacher to upload audio, video and texts based content and integrate into customized study modules which should enable a multi teacher access facility for sharing and collaborating content resources remotely across the campus.	
		The teacher should be able to create session modules for students access a content management library on the media activity dashboard for self-study even when the teacher is not present in class.	
		The system should facilitate teacher to create different group and perform group discussion or pair discussion at the same time for different group and record their discussion.	
2.	Model Limitation	Students listen to and repeat after a model track to practice the stress rhythm and intonation of a foreign language, while their voices are recorded onto the student track. You can either use sources with readymade pauses in them or manually pause the audio source during the activity.	

Ser No	Specification	Description	Remarks
3.	Round Table Discussion	Round Table discussion allows teachers to set up small student groups with a designated chairman, who leads the discussion and assigns turns during the discussion by passing a microphone around the group. The activity allows teachers to set up a meeting environment to help advanced language learners strengthen their oral presentation skills. Chairmen and participants for the groups can be defined either manually or with automatic options. Student applications display a list of discussion participants and chairman and students can request the microphone by clicking a request button. Teacher can listen in on each group and participate in the discussion when needed.	
4.	Content Exercise	The content exercise activity is used together with either readymade or custom content provided by reputed publisher. The activity allows you to launch content files directly to students into an exercise window that includes all text, image and audio resources for the exercise. Report tool for the content exercise activity should allow teachers to view, edit and save a detailed listing of student scores. Content should be HTML format and grade wise. English content should be known publisher link Encyclopedia Britannica. Content should be integrated in the language lab software and level wise.	
5.	Vocabulary Test	The activity allows teachers to easily conduct word quiz-type language learning tests. The teacher can either enter the quiz words manually or by importing a ready word list and students will then need to simply fill in their answers in a dialog that is launched on their screens. The activity allows teachers to create, launch and save tests, as well as collect result. Teachers are also able to display individual results to students and mark answers as correct or incorrect.	
6.		Telephone activity allows students to call each other, which lets you to perform a wide range of real-life based conversation exercises and to improve student engagement during discussion exercises.	
7.	Pronunciation Activity	The pronunciation activity provides an in-built pronunciation tool for teachers and students by integrating the pronounce solution into a class room. The pronunciation activity allows teachers to convert any text into a pronunciation exercise that can be launched directly to students. At the student end, a separate pronunciation application will open where they can proceed with the pronunciation exercise.	
8.		Additional Module like conference interpreting activity provides an authentic interpretation training environment. The activity simulates a real-life meeting with a live discussion that is simultaneously interpreted to different languages. Students can be assigned as Interpreters, Delegates and Observers. During the activity users are able to select whether to listen to the discussion or to an interpreter and interpretation recordings can be collect afterwards.	
9.		Teachers have the possibility to record full presentation with screen capture and audio. This allows students to practice giving presentations and to review their work, or teachers to record parts of their lesson for later use.	

Ser No	Specification	Description	Remarks
10.		English Content Should contain: Platform should have skill based interactive software listening, speaking, reading, grammar and better communication. Pronunciation. The international Phonemic alphabet, Similar sounds gives extensive practice in recognizing and distinguishing similar-sounding letters (minimal-pairs) which can be problematic. The game-style format is very popular with learners word and phrasal stress helps learners to recognize where the stress falls in words of more than one syllable, and in short phrases. This is done through a variety of fun exercise types. Stress and Rhythm section containing over 600 sound items, introduces learners to the features of connected speech, including the rules governing sentencing stress, weak forms, linking, elision, assimilation and contraction, and shifting stress. It takes the form of a step by step tutorial with test-as-you-learn features and extensive practice in the focus areas. It also contains a recording tool and back chaining facility. Rhythm from Rain Land. Practice in the rhythm of spoken English throughout everyday dialogues, jokes, poems, rhymes and quotations the practice takes the form of video and audio lessons where the learner can listen and repeat the language in the texts in a variety of ways. There are 48 texts which you may access unit-by-unit or by topic area. The program also has a special recording feature, which allows the learner to build up a set of recordings. This is the program of choice for British Council (global license), international schools and hundreds of universities, colleges and schools worldwide. A personal progress page to chart scores and assess progress tailored learning capability where the teachers can create a special course by selecting the individual exercise to be completed by their students, set the hours to be allocated to the course and specify the score for students to attain. Read Up-Speed Up. Improve reading skills in students. Active Listening. A listening for information task- here the learner must listen for something specific e.g. what food items are mentioned by a shopper. A listen and Respond Task - here the learners must take the part of one of the speakers and respond by choosing the most suitable option. Areas covered include meeting people, eating out, and booking tickets. Listening and Note-taking tasks - where the listener must fill out forms, take message and complete timetables etc. A General Listening Task - here the learner listen to an interview or to a talk and must first answer gist questions and then specific questions on the text.	

Ser No	Specification	Description	Remarks
		- Result Tracking. Each time they do the test the learner's scores are recorded so they can see their scores from all previous tests and check their progress. Teachers can access these scores to assess learner. Progress and help learners to pinpoint areas to review and work on. - Follow up quiz. To check learner's comprehension- the learner can access a larger grammar section which highlights the particular areas where the learner mistake. The learner can then study these areas. Each tense covered in the grammar section has a quiz to check the learner's comprehension. - Large Test Bank. Means that each time the learner does the test they will get a different set of questions. This means that the learner can do the test and again and again and can continue to trust the validity of the test. Branded Publisher content for school should be integrated with Language Lab Software. Content Worksheet should include different exercise covering all four key skills, reading, writing, speaking and listening should follow CEFR.	
11	Demo	Demo if reqd by the customer should be done and each above functionality should be shown.	

Annexure-II

MINIMUM REQUIRED TECHNICAL SPECIFICATIONS : FOR PROCUREMENT & INSTALLATION OF ONE DIGITAL LANGUAGE LAB WITH COMPLETE ACCESSORIES

TECHNICAL SPECIFICATION FOR HEAD SET WITH MIC			
S No	Specification	Description	Remarks
1.	Specification	Headphone - Frequency Response: 10Hz – 21kHz - Sensitivity : 100dBSPL/mW at 1kHz - Total Harmonic distortion : < 2% Microphone - 10mm Electret Condenser, noise cancelling, unidirectional with built in mesh filter. - Frequency Response :50 Hz - 17 kHz - Sensitivity : 42 dBV (0dB = 1V/Pa at 1kHz) Comfort and design - Form factor : Over-ear, circumaural, closed back. - Ear cushion : HyperX signature memory foam wrapped in premium leatherette. - Headband : Memory foam with soft leatherette covering. - Frame : Full Metal frame for durability and flexibility. Connectivity - Wired Connection : 3.5mm (04 pole CTIA), USB-A, USB-C. - USB Specification : USB 2.0 & Stereo audio format. - Cable Length : 1.2m Headset cable, 1.3m USB dongle cable. - Sampling Rates : 44.1 kHz, 48 kHz and 96 kHz. - Bit Depth : 16 bit , 24 bit. - Audio Control : On board volume and mute.	

Annexure-III

MINIMUM REQUIRED TECHNICAL SPECIFICATIONS : FOR PROCUREMENT & INSTALLATION OF ONE DIGITAL LANGUAGE LAB WITH COMPLETE ACCESSORIES

TECHNICAL SPECIFICATION FOR DESKTOP PC ALL-IN-ONE			
S No	Specification	Description	Compliance Yes/No
1.	Processor	CPU- Intel® Core™ i7-13 th Gen or higher (1.5 GHz E-Core base frequency, 2.1 GHz P-core base frequency, upto 4.2 GHz E-core Max Turbo frequency, up to 5.3 GHz P-core Max Turbo frequency, 33 MB L3 cache, 8 P-cores and 12 E-cores, 28 threads) with Intel® UHD Graphics 770	
2.	Display	Display 23.8" FHD	
3.	Memory	16 GB DDR4 /DDR5, maximum memory -64 GB DDR-4-3200 SDRAM, 64 GB DDR5-5600 SDRAM, Transfer rates upto 3200 MT/s: Transfer rates upto 5600 MT/s	
4.	Storage	512 GB NVMe™ M.2 SSD	
5.	Operating System	Windows 11 Professional (pre-loaded) with Audio Visual Tools	
6.	Audio	Realtek ALC3687 Codec, universal Audio Jack with CTIA headset support, audio line-out rear ports (3.5mm), multi-streaming capable	
7.	Communication	LAN: Integrated 10/100/1000M GbE	
8.	Ports and Connectors	2 USB Type-A 10 Gbps signaling rate, 2USB Type-A 5Gbps signaling rate, 1 USB Type-C®, 1 USB 2.0 Type-A, 1 headphone/microphone combo, Rear 2 USB 2.0 Type-A, 1 HDMI, 1 Display Port™ 1RJ-45, 1 Power connector, 1 serial, 1 line in, 1 line out	
9.	Warranty	Onsite site from OEM 3-3-3 years	

Annexure-IV

MINIMUM REQUIRED TECHNICAL SPECIFICATIONS : FOR PROCUREMENT & INSTALLATION OF ONE DIGITAL LANGUAGE LAB WITH COMPLETE ACCESSORIES

TECHNICAL SPECIFICATION FOR ONLINE UPS			
S No	Specifications	Specifications / Features	Compliance Yes/No
1.	Specifications	Power- 20 KVA 1 phase 220/230/240 V ; 3 phase 380/400/415 V configurations UPS	Yes/No
Input			
2.	Rated Input Voltage	1 phase 220/230/240 V ; 3 phase 380/400/415V	
3.	Rated Voltage Range	160 V ~ 300V full load; 100-160V Liner derating	
4.	Rated Input Frequency	50 Hz/ 60 Hz	
5.	Input Frequency Range	40 Hz ~ 70 Hz	
6.	Input Frequency phase lock range	50Hz System; 45 Hz-55Hz: 60Hz system 54Hz-66Hz	
7.	System Compatibility	TN-S / IT	
8.	Input Power factor (PF)	>0.995 both 1 phase and 3phases	
Output			
9.	Input output phase connection	Input-output 3-1,3-3,1-1	
10.	Rated output voltage	1 phase 220/230/240 V ; 3 phase 380/400/415V	
11.	Rated output Frequency	50 Hz/ 60 Hz	
12.	Rated output appearance Frequency	20 KVA	
13.	Max PF	1	
14.	Voltage variation	± 1%	
15.	Load crest ratio	>=3:1	
16.	Output connection	Terminal block	
Battery and charger			
17.	Internal Batteries	None : 2x EBM's a minimum to provide backup time.	
18.	Max Quantity	6 Pairs (2*6)	
19.	Battery Voltage	± 240V (adjustable to ± 192V, use with correct battery)	
20.	Recharging time	3 hours to 90% (2 EBM)	
Certification			
21.	Safety	CE/TLC/RCM	
22.	Energy Saving	CQC	
Battery			
23.	Battery	12V/100AH/ EXIDE SMF Battery Warranty - 2 years for 60 minutes backup	
Rack & Interconnectors			
24.	Rack & Interconnectors	1 Set	

Annexure-V

MINIMUM REQUIRED TECHNICAL SPECIFICATIONS : FOR PROCUREMENT & INSTALLATION OF ONE DIGITAL LANGUAGE LAB WITH COMPLETE ACCESSORIES

TECHNICAL SPECIFICATION FOR NETWORKING COMPONENTS			
S No	Specifications	Specifications / Features	Compliance Yes/No
1.	Specifications	24 Port Gigabit-Data Switch 2 Semi Manage with 2 SFP- 2 Nos	
2.	Patch Panel	24-port-patch panel for managing the networking wires - 2 Nos	
3.	Input / Output Box	Complete IO box with 1 x keystone Jack, 1 x Gang Box, 1 x Socket Face Plate -42 Nos	
4.	Networking accessories	1 Mtr CAT 6 Patch cord - 45 Nos	
5.		2 Mtr CAT 6 Patch cord - 42 Nos	
6.		CAT 6 cables - Cat 6 Cables 23AWG (2 Box x 305 MT) – 02 Nos	
7.	Installation and commissioning		